

Structure in Attitude Reports

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1 Recap and matters arising

The paper discussed last week defended the claim that the following schema is *valid*, so that its instances are true on all *uniform* resolutions of their context-sensitivity:

ATTITUDE TRANSPARENCY For any things x and y : if $x = N$, then y As that $\phi[x]$ iff y As that $\phi[N]$.

(Despite the Loglish, this is supposed to be a schema whose instances are English sentences.) This is intuitive in itself, but has the less-intuitive implication that the following schema is also valid:

NAME SUBSTITUTION [For any y :] If $N' = N$, then y As that $\phi[N']$ iff y As that $\phi[N]$.

Here, N and N' are restricted to *ordinary proper names*—not definite descriptions, or complex expressions like ‘five plus seven’ or ‘[the property of] being an atom with one proton’....

- If one replaces N with a definite description like ‘the shortest spy’, the resulting sentence is plausibly *structurally ambiguous* between a valid parsing (‘de re’) and an invalid one (‘de dicto’). (Is this also true for ‘five plus seven’ and ‘being an atom with one proton’?)

1.1 Nonstandard contexts

One surprising consequence of the picture is that false-seeming sentences like the following sentences admit true interpretations:

- (1) Lois knows that Clark flies.
- (2) Lois doesn’t know that Superman flies.

This implication is defensible quite apart from NAME SUBSTITUTION (consider conversations at gatherings of superheroes where everyone is on familiar terms).

1.2 Multiple names inside one attitude verb

- (3) Plato didn’t know that Hesperus is identical to Phosphorus.

Should we say that (3) (i) has some true uniform interpretations; (ii) has some true non-uniform interpretations but no true uniform ones; or (iii) doesn’t even have any true non-uniform interpretations?

Well, given the validity of NAME SUBSTITUTION, whatever we say about (3) we will also have to say about

- (4) Plato didn’t know that Hesperus is identical to Hesperus.

I think (ii) seems like the best compromise.

But this raises a puzzle: *which elements* are being interpreted nonuniformly when we get the true readings of these sentences?

2 Some more controversial claims about the logic of attitudes

2.1 Closure rules

Suppose we have a syntactic operation O that turns formulae in some language into other formulae, and consequence relations $\vdash_1$ and $\vdash_2$ (maybe identical) on the formulae in that language. Here are three salient conditions on these things.

CLOSURE If $P_1, P_2 \dots \vdash_1 Q$, then $OP_1, OP_2 \dots \vdash_2 OQ$.

MONOTONICITY If $P \vdash_1 Q$, then $OP \vdash_2 OQ$.

CONGRUENTIALITY If $P \vdash_1 Q$ and $Q \vdash_1 P$ then $OP \vdash_2 OQ$.

In the tradition of epistemic and doxastic logic, the operations that turn P into ‘ A believes that P ’ and ‘ A knows that P ’ (where A is a singular term) are widely taken to have these properties—at least for some $\vdash_1$ including some basic logic (e.g. classical propositional logic) and some $\vdash_2$ that implies *preserving truth on all uniform interpretations*.¹

Note that assuming $\vdash_1$ respects the usual rules for ‘and’, Congruentiality implies Monotonicity given

AND-OUT $O(P \text{ and } Q) \vdash_1 OP$

and Monotonicity implies Closure (for finite premise sets) given

AND-IN $OP, OQ \vdash_1 O(P \text{ and } Q)$

¹At least if we add something like ‘ A is an agent’/‘ A believes something’/‘ A knows something’ as an extra side premise.

There’s also a tradition of acknowledging that this might involve some departure from the usual understanding of ‘validity’:

Hence there need not be anything nonsensical, irrational, or dishonest about a set of sentences which I have called inconsistent even when they are uttered by one and the same person on one and the same occasion.... They may even be true simultaneously. This does not go to show that our rules are incorrect, however. What it shows is that the notion which they define is unlike inconsistency in the current senses of the word, and should be carefully distinguished from it. (HintikkaKB)

There's a very widely-used semantic theory of 'believe' and 'know' (often attributed, I think inaccurately, to **HintikkaKB**) that implies Closure for these verbs:

$$\begin{aligned} \llbracket \text{believes } P \rrbracket(y, w) &= 1 \text{ iff } \llbracket P \rrbracket(v) = 1 \text{ for all } v \in \text{Dox}(y, w) \\ \llbracket \text{knows } P \rrbracket(y, w) &= 1 \text{ iff } \llbracket P \rrbracket(v) = 1 \text{ for all } v \in \text{Epi}(y, w) \end{aligned}$$

where *Dox* and *Epi* are certain functions that take an individual and a world to a set of worlds.²

No-one is so misguided that *all* attitude verbs/adjectives obey Closure and Monotonicity. But giving up the view that they all obey Congruentiality seems to require a fairly radical overhaul of the whole framework of "possible worlds semantics".

2.2 Generalized substitution

There are various expressions in English which one might think of as analogues of identity for other grammatical categories:

- To be a tadpole is to be a juvenile frog.
- Kicking is the same as hitting with one's hand.
- The property of being a hydrogen atom is identical to the property of being an atom with one proton.
- For there to be a tadpole in the bowl is for there to be a juvenile frog in the bowl.

Insofar as one thinks of these as identity-like, it's natural to suppose that if Name Substitution is valid, so is Generalized Substitution:

GENERALIZED SUBSTITUTION If $F = G$, then y As that $\phi[F]$ iff y As that $\phi[G]$.

where ' $F = G$ ' stands for the appropriate kind of generalized identity given the syntactic category of ' F ' and ' G '.

It's not clear whether there are *intuitive* motivations for this comparable to the argument for NAME SUBSTITUTION from ATTITUDE TRANSPARENCY. But there are *theoretical* motivations for it, comparable to the argument for NAME SUBSTITUTION from the Millian view that there is nothing more to the meaning of a name than its referent. One might by analogy think that there is nothing more than the meaning of a predicate than what *property* it refers to; and that the truth of 'To be F is to be G ' requires (the relevant uses of) F and G to refer to the same property.

One could argue for some form of Congruentiality from the combination of GENERALIZED SUBSTITUTION with the metaphysical view that equivalence in the sense of

$\vdash_1$ implies higher-order identity. Such metaphysical views are controversial but in many cases attractive:

- To be a tadpole is to not not be a tadpole.
- To be a tadpole is to either be a tadpole or a tadpole.

I find this to be the most interesting kind of consideration in favour of Congruentiality.

3 Some putative counterexamples

Let's start with some putative counterexamples to Generalized Substitution:

- (5) a. The kindergartners know that there is a tadpole in the bowl.
b. The kindergartners don't know that there is a juvenile frog in the bowl.
c. To be a tadpole is to be a juvenile frog.
- (6) a. Lavoisier knew that every hydrogen atom is a hydrogen atom.
b. Lavoisier didn't know that every hydrogen atom has exactly one proton.
c. To be a hydrogen atom is to have exactly one proton.
- (7) a. I forgot that every semigroup is associative.
b. I didn't forget that every associative magma is associative.
c. To be a semigroup is to be an associative magma.

(You might think that these cases should be treated as *reductios* of the higher-order identities in question. But I think this is a bad idea.)

Putative counterexamples to Congruentiality based on imperfect logical acumen:

- (8) a. So-and-so knows that every barber is a barber.
b. So-and-so doesn't know that there is no village in which some person is a barber who shaves every person in that village who does not shave himself.
- (9) a. Dedekind knows that zero is not a successor, and no two numbers share a successor, ...and any property had by zero and by the successor of all its instances is had by all numbers.
b. Dedekind does not know that [...and] no number greater than the successor of the successor of zero is such that the sum of two numbers raised to that power is some number raised to that power.
- (10) a. So-and-so thinks that there is a village in which some person is a barber who shaves every person in that village who does not shave himself.

²Or equivalently: three-place relations between an individual and two worlds.

- b. So-and-so does not think that some barber is not a barber.

Putative counterexamples to Congruentiality (and maybe Generalized Substitution) based on logical or philosophical error:

- (11) a. Crispin believes that Harry is both bald and not bald.
b. Crispin does not believe that Harry is either bald or not bald.

Putative counterexamples based on unawareness:

- (12) a. Alexander believed that any war with Napoleon would be disastrous.
b. Alexander believed that any nuclear war with Napoleon would be disastrous and any non-nuclear war with Napoleon would be disastrous.

I'm particularly interested in putative counterexamples based on 'double vision':

- (13) a. Peter believes that Paderewski is a politician and Paderewski plays the piano (**KripkePAB**).
b. Peter does not believe that Paderewski is a politician who plays the piano.
(14) a. [Some man]₁ on the beach is such that for [some man]₂ on TV, Ralph believes that he₁ is a spy and he₂ works for the government
b. It's not the case that [some man]₁ on the beach is such that for [some man]₂ on TV, Ralph believes that he₁ is a spy who works for the government.

4 Contextualism to the rescue?

Three contextualist moves to defend the validity of an argument from a true-seeming premise to a false-seeming conclusion.

- (i) The premise has both true and false uniform interpretations, as does the conclusion. But the true ones are salient for the premise and the false ones are salient for the conclusion.
(ii) The premise has salient true interpretations, but these are all nonuniform. The conclusion has no true uniform interpretations, and maybe no true interpretations at all.
(iii) The conclusion has salient false interpretations, but these are all nonuniform. The premise has no false uniform interpretations, and maybe no false interpretations at all.

Could we defend Congruentiality and/or Generalized Substitution from the foregoing counterexamples in one of these ways?

The obvious worry about options (i) and (iii) is that the false-seeming conclusion seems so bad that it's not plausible that it admits *any* true interpretations, or at least any uniform ones.

- But it would also be enough if one thought that the conclusion was structurally ambiguous between several readings where *one* of the readings lacks true uniform interpretations, provided the premise doesn't have a parallel reading.

Option (ii) might seem tempting for the 'double vision' cases. The most obvious way of developing this view would also say that sentences like the following lack true uniform interpretations:

- (15) Peter believes that Paderewski is a politician and Peter believes that Paderewski plays the piano.
(16) Someone on the beach is such that Ralph believes he is a spy, and someone on TV is such that Ralph believes he is not a spy.

But there is a strong argument, sort of implicit in **KripkePAB**, that (15) has true uniform interpretations. Imagine that there are two other people X and Y each of whom only knows Paderewski in one of the two relevant ways. Then there seems to be a true uniform reading of

- (17) X believes that Paderewski is a politician and Y believes that Paderewski plays the piano.

But the following seems to be true on *all* uniform interpretations:

- (18) If X believes that Paderewski is a politician, Peter does, and if Y believes that Paderewski plays the piano, Peter does.

Given this, there must be true uniform readings of (15) too. A parallel argument can be made for (16).

There are also subtler ways of developing option (ii) for the 'double vision' cases, on which one accepts that (15)/(16) has true uniform interpretations, but denies that this carries over to other sentences such as the following:

- (B) There is someone such that Ralph believes he is a spy and believes he is not a spy. ($\exists x.B(Sx) \wedge B(\neg Sx)$)
(B') There is someone₁ and there is someone₂ identical to him₁ such that Ralph believes that he₁ is a spy and he₂ is not a spy. ($\exists xy.x = y \wedge B(Sx \wedge \neg Sy)$)
(C) There is someone such that Ralph believes he is a spy and he is not a spy. ($\exists x.B(Sx \wedge \neg Sx)$)

Kaplan's view: each occurrence of an attitude verb introduces one existential quantifier over guises ('vivid modes of presentation') for each variable free in its scope. So C is false, while B and B' are true. Note that this involves a failure of the analogue of Attitude Transparency for variables!

Salmon/Soames view: distinct *occurrences* of the same name or variable correspond

to different existential quantifiers over guises. So B, B', and C are all true, even though

(D) There is someone such that Ralph believes he is a spy who isn't a spy.
 $(\exists x.B((S \wedge \neg S)x))$

is false.

Percus/Sauerland view: the sentences B' and C are *structurally ambiguous*. The two pronouns can either be 'co-ordinated' (associated with the same guise generator variable) or 'un-coordinated'. On the co-ordinated interpretation, the sentences are impossible (at least given Ralph's logical competence); on the un-coordinated interpretation, they have true (and uniform) interpretations.

		Stalnaker	Williamson	G&L	Kaplan	Soames	me	CGL?	Aloni	P&S
A	$\exists xy.x = y \wedge B(Sx) \wedge B(\neg Sy)$	×	✓	?	✓	✓	?	✓	?×	?
B	$\exists x.B(Sx) \wedge B(\neg Sx)$	×	✓	?	✓	✓	?	×	×	?
B'	$\exists xy.x = y \wedge B(Sx \wedge \neg Sy)$	×	✓	?	✓	✓	?	✓	?×	?×
C	$\exists x.B(Sx \wedge \neg Sx)$	×	✓	?	×	✓	?	×	×	?×
D	$\exists x.B((S \wedge \neg S)x)$	×	✓	?	×	×	×	×	×	×

5 A fallback

Even if we deny that GENERALIZED SUBSTITUTION is valid, we might accept its restriction to *semantically atomic* expressions. Call this 'Atomic Substitution'.

None of the previous counterexamples threaten this. (And they don't even threaten the version that replaces the generalized identity with a claim of metaphysically necessary equivalence.)

One might think that the following is a counterexample:

- (19) a. Everyone knows that all attorneys are attorneys.
 b. Not everyone knows that all attorneys are lawyers.
 c. To be an attorney is to be a lawyer.

But it seems very natural to say the same thing here that we say about 'Hesperus'/'Phosphorus'.

Let's consider some views about the truth-conditions of belief sentences that can

accept Atomic Substitution but not Generalized Substitution. (Note: we are not giving semantic *implementations* here.)

5.1 Mentalese isomorphs

'C believes that $R(A, B)$ ' is necessarily equivalent to

- (20) C's belief box contains a sentence formed by composing a (contextually-relevant) predicate expressing R with a (contextually-relevant) name referring to A and a (contextually-relevant) name referring to B .

More generally: for each occurrence of a semantically atomic constituent under 'believe', we get a separate existential quantifier over contextually-relevant expressions in the subject's mentalese referring to the item in question; these are required to be assembled in a way that's isomorphic to the prejacent.

5.2 Descriptivism

'C believes that $R(A, B)$ ' is necessarily equivalent to

- (21) For some contextually-relevant properties F, G, H had uniquely by R, A , and B, C doxizes the state of affairs that the G object and the H object stand in the F relation.

where 'doxizing' is some underlying relation between individuals and states of affairs (which could be identified with sets of possible worlds).

More generally: for each for each occurrence of a semantically atomic constituent under 'believe', we get a separate existential quantifier over contextually-relevant properties had uniquely by the item in question: the subject is required to doxize a state of affairs got by chunking these together in a way that's isomorphic to the prejacent.